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United States Patent	12409979
Kind Code	B2
Date of Patent	September 09, 2025
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System and method for implementing cap closure for carbonated and oxygen sensitive beverages

Abstract

A system and method for implementing cap closure for a carbonated beverage is disclosed. According to one embodiment, an apparatus includes a cap liner having a circular ring shape. The apparatus further comprises an outer lip and an inner portion of the circular ring shape. The outer lip is wider than the inner portion, and the outer lip has two or more members extending away from a center of the outer lip.

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Appl. No.: 17/066252

Filed: October 08, 2020

Prior Publication Data

Document Identifier	Publication Date
US 20210163186 A1	Jun. 03, 2021

Related U.S. Application Data

continuation parent-doc US 14608016 20150128 US 10815035 child-doc US 17066252
us-provisional-application US 61932701 20140128

Publication Classification

Int. Cl.: B65D41/14 (20060101); B65D41/04 (20060101)

U.S. Cl.:

CPC B65D41/145 (20130101); B65D41/045 (20130101);

Field of Classification Search

CPC: B65D (41/045); B65D (7/36); B65D (43/0212); B65D (1/0253); B65D (1/0246)

USPC: 215/341; 215/349

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application for patent claims the benefit of and priority to and is a continuation of U.S. application Ser. No. 14/608,016, filed Jan. 28, 2015, entitled “System and Method for Implementing Cap Closure for Carbonated and Oxygen Sensitive Beverages” which claims priority to U.S. Provisional Patent Application No. 61/932,701, filed on Jan. 28, 2014, entitled “System and Method for Implementing Cap Closure for Carbonated Wine”, which is herein incorporated by reference.

FIELD

(1) The present disclosure relates in general to cap closures. In particular, the present disclosure relates to a system and method for implementing cap closure for carbonated and oxygen sensitive beverages such as wine.

BACKGROUND

(2) A number of wineries offer lower alcohol (e.g. 9%), lightly sweet, semi-sparkling wines. These semi-sparkling wines are marketed and promoted to consumers as casual and approachable wines that are ideal for outdoor get-togethers with family and friends. Semi-sparkling wine contains a significant level of carbon dioxide that gives its fizzy appearance and effervescent mouth feel. The

pressure in a bottle of semi-sparkling wine typically varies from approximately 0.3 to 2 atmospheres which equates to concentrations of 2 to 5 g CO₂/L at 20° C.

(3) There are four main issues when applying traditional wine closures for these semi-sparkling wines: brand image, ease of opening, re-sealability, and pressure retention. Champagne corks with wire hoods are too formal. Crown closures have a more relaxed image, but are not easy to open. Additionally, both champagne corks and crown closures are not designed to be easily reapplied to the package by the consumer. Long-skirt screw-caps (e.g. 30 mm diameter×60 mm tall aluminum closures with traditional SARANEX™ liners) provide the right marketing image and are easy to re-apply. However, lab tests show that these long-skirt screw-caps containing SARANEX liners cannot consistently retain an internal pressure greater than 40 psi within a bottle of semi-sparkling wine. Such internal pressures in these semi-sparkling wines can be reached under typical shipping and storage conditions.

(4) There are several injection molded liner technologies commercially available for carbonated wine such as GUALA® MOSS and ERBEN® ASTRO. These liners are specifically designed for carbonated products and are typically used in 30×60 mm aluminum screw-caps. These liners are made of a low-density polyethylene material resulting in relatively low material costs but comparatively high oxygen transmission rates (OTR) of 0.003 cc O₂/closure/24 hours.

SUMMARY

(5) A system and method for implementing cap closure for a carbonated beverage is disclosed. According to one embodiment, an apparatus includes a cap liner having a circular ring shape. The apparatus further comprises an outer lip and an inner portion of the circular ring shape. The outer lip is wider than the inner portion, and the outer lip has two or more members extending away from a center of the outer lip.

(6) The above and other preferred features, including various novel details of implementation and combination of elements, will now be more particularly described with reference to the accompanying figures and pointed out in the claims. It will be understood that the particular systems and methods described herein are shown by way of illustration only and not as limitations. As will be understood by those skilled in the art, the principles and features described herein may be employed in various and numerous embodiments.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) The accompanying figures, which are included as part of the present specification, illustrate the various embodiments of the present disclosed system and method and together with the general description given above and the detailed description of the preferred embodiments given below serve to explain and teach the principles of the present disclosure.

(2) FIG. 1(a) illustrates a top view of an exemplary cap liner, according to one embodiment.

(3) FIG. 1(b) illustrates a cross-sectional view of an exemplary cap liner as illustrated in FIG. 1(a), according to one embodiment.

(4) FIG. 1(c) illustrates a detailed cross-sectional view of an exemplary cap liner as illustrated in FIG. 1(b), according to one embodiment.

(5) FIG. 2(a) illustrates a top view of another exemplary cap liner, according to one embodiment.

(6) FIG. 2(b) illustrates a cross-sectional view of another exemplary cap liner as illustrated in FIG. 2(a), according to one embodiment.

(7) FIG. 2(c) illustrates a detailed cross-sectional view of another exemplary cap liner as illustrated in FIG. 2(b), according to one embodiment.

(8) FIG. 3(a) illustrates a top view of another exemplary cap liner, according to one embodiment.

(9) FIG. 3(b) illustrates a cross-sectional view of another exemplary cap liner as illustrated in FIG.

3(a), according to one embodiment.

(10) FIG. 3(c) illustrates a detailed cross-sectional view of another exemplary cap liner as illustrated in FIG. 3(b), according to one embodiment.

(11) FIG. 4(a) illustrates a top view of another exemplary cap liner, according to one embodiment.

(12) FIG. 4(b) illustrates a cross-sectional view of another cap liner as illustrated in FIG. 4(a), according to one embodiment.

(13) FIG. 4(c) illustrates a detailed cross-sectional view of another exemplary cap liner as illustrated in FIG. 4(b), according to one embodiment.

(14) FIG. 5(a) illustrates a top view of another exemplary cap liner, according to one embodiment.

(15) FIG. 5(b) illustrates a cross-sectional view of another exemplary cap liner as illustrated in FIG. 5(a), according to one embodiment.

(16) FIG. 5(c) illustrates a detailed cross-sectional view of another exemplary cap liner as illustrated in FIG. 5(b), according to one embodiment.

(17) FIG. 6(a) illustrates a top view of another exemplary cap liner, according to one embodiment.

(18) FIG. 6(b) illustrates a cross-sectional view of another cap liner as illustrated in FIG. 6(a), according to one embodiment.

(19) FIG. 6(c) illustrates a detailed cross-sectional view of another exemplary cap liner as illustrated in FIG. 6(b), according to one embodiment.

(20) FIG. 7(a) illustrates a top view of another exemplary cap liner, according to one embodiment.

(21) FIG. 7(b) illustrates a cross-sectional view of another exemplary cap liner as illustrated in FIG. 7(a), according to one embodiment.

(22) FIG. 7(c) illustrates a detailed cross-sectional view of another exemplary cap liner as illustrated in FIG. 7(b), according to one embodiment.

(23) FIG. 8(a) illustrates a top view of another exemplary cap liner, according to one embodiment.

(24) FIG. 8(b) illustrates a cross-sectional view of another exemplary cap liner as illustrated in FIG. 8(a), according to one embodiment.

(25) FIG. 8(c) illustrates a detailed cross-sectional view of another exemplary cap liner as illustrated in FIG. 8(b), according to one embodiment.

(26) FIG. 9(a) illustrates a cross-sectional view of another exemplary cap liner in a screw cap, according to one embodiment.

(27) FIG. 9(b) illustrates a detailed cross-sectional view of another exemplary cap liner in a screw cap as illustrated in FIG. 9(a), according to one embodiment.

(28) FIG. 9(c) illustrates a top view of another exemplary cap liner as illustrated in FIG. 9(a), according to one embodiment.

(29) FIG. 9(d) illustrates a cross-sectional view of another exemplary cap liner as illustrated in FIG. 9(c), according to one embodiment.

(30) FIG. 9(e) illustrates a detailed cross-sectional view of another exemplary cap liner as illustrated in FIG. 9(d), according to one embodiment.

(31) FIG. 10(a) illustrates a detailed cross-section of an outer lip diagram showing how outer lip designs were made to conduct a design-of-experiment on liner outer lip structures, according to one embodiment.

(32) FIG. 10(b) illustrates a design-of-experiment array created to determine the effect of various structural elements on the liner performance characteristics, according to one embodiment.

(33) FIG. 11(a) illustrates a top view of another exemplary cap liner, according to one embodiment.

(34) FIG. 11(b) illustrates a cross-sectional view of another exemplary cap liner as illustrated in FIG. 11(a), according to one embodiment.

(35) FIG. 11(c) illustrates a detailed cross-sectional view of another exemplary cap liner as illustrated in FIG. 11(b), according to one embodiment.

(36) FIG. 12(a) illustrates a top view of another exemplary cap liner, according to one embodiment.

(37) FIG. 12(b) illustrates a cross-sectional view of another exemplary cap liner as illustrated in

FIG. 12(a), according to one embodiment.

(38) FIG. 12(c) illustrates a detailed cross-sectional view of another exemplary cap liner as illustrated in FIG. 12(b), according to one embodiment.

(39) FIG. 13(a) illustrates a top view of another exemplary cap liner according to one embodiment.

(40) FIG. 13(b) illustrates a cross-sectional view of another exemplary cap liner as illustrated in FIG. 13(a), according to one embodiment.

(41) FIG. 13(c) illustrates a detailed cross-sectional view of another exemplary cap liner as illustrated in FIG. 13(b), according to one embodiment.

(42) FIG. 14(a) illustrates a top view of another exemplary cap liner according to one embodiment.

(43) FIG. 14(b) illustrates a cross-sectional view of another exemplary cap liner as illustrated in FIG. 14(a), according to one embodiment.

(44) FIG. 14(c) illustrates a detailed cross-sectional view of another exemplary cap liner as illustrated in FIG. 14(b), according to one embodiment.

(45) FIG. 15(a) illustrates a top view of another exemplary cap liner according to one embodiment.

(46) FIG. 15(b) illustrates a cross-sectional view of another exemplary cap liner as illustrated in FIG. 15(a) according to one embodiment.

(47) FIG. 15(c) illustrates a detailed cross-sectional view of another exemplary cap liner as illustrated in FIG. 15(b), according to one embodiment.

(48) FIG. 16(a) illustrates a top view of another exemplary cap liner according to one embodiment.

(49) FIG. 16(b) illustrates a cross-sectional view of another exemplary cap liner as illustrated in FIG. 16(a), according to one embodiment.

(50) FIG. 16(c) illustrates a detailed cross-sectional view of another exemplary cap liner as illustrated in FIG. 16(b), according to one embodiment.

(51) FIG. 17(a) illustrates a top view of another exemplary cap liner according to one embodiment.

(52) FIG. 17(b) illustrates a cross-sectional view of another exemplary cap liner as illustrated in FIG. 17(a), according to one embodiment.

(53) FIG. 17(c) illustrates a detailed cross-sectional view of another exemplary cap liner as illustrated in FIG. 17(b), according to one embodiment.

(54) FIG. 18(a) illustrates a top view of another exemplary cap liner according to one embodiment.

(55) FIG. 18(b) illustrates a cross-sectional view of another exemplary cap liner as illustrated in FIG. 18(a), according to one embodiment.

(56) FIG. 18(c) illustrates a detailed cross-sectional view of another exemplary cap liner as illustrated in FIG. 18(b), according to one embodiment.

(57) FIG. 19(a) illustrates a top view of another exemplary cap liner according to one embodiment.

(58) FIG. 19(b) illustrates a cross-sectional view of another exemplary cap liner as illustrated in FIG. 19(a), according to one embodiment.

(59) FIG. 19(c) illustrates a detailed cross-sectional view of another exemplary cap liner as illustrated in FIG. 19(b), according to one embodiment.

(60) FIG. 20(a) illustrates a cross-sectional view of another exemplary cap liner in a screw cap, according to one embodiment.

(61) FIG. 20(b) illustrates a detailed cross-sectional view of another exemplary cap liner in a screw cap as illustrated in FIG. 20(a), according to one embodiment.

(62) FIG. 20(c) illustrates a top view of another exemplary cap liner according to one embodiment.

(63) FIG. 20(d) illustrates a 3-dimensional view of another exemplary cap liner as illustrated in FIG. 20(c), according to one embodiment.

(64) FIG. 20(e) illustrates a cross-sectional view of another exemplary cap liner as illustrated in FIG. 20(c), according to one embodiment.

(65) FIG. 20(f) illustrates a detailed cross-sectional view of another exemplary cap liner as illustrated in FIG. 20(e), according to one embodiment.

(66) FIG. 21 illustrates an exemplary graph for determining a maximum internal package pressure

in a rigid container filled to the appropriate volume, according to one embodiment.

(67) FIG. 22 illustrates an exemplary plot of the effect of various types of cap liners on slip torque, according to one embodiment.

(68) FIG. 23 illustrates an exemplary plot of the effect of various types of cap liners on break torque, according to one embodiment.

(69) FIG. 24 illustrates an exemplary plot of the effect of various types of cap liners on secure seal test (SST), according to one embodiment.

(70) FIG. 25 illustrates an exemplary plot of the effect of various types of cap liners on oxygen transmission rate (OTR), according to one embodiment.

(71) FIG. 26 illustrates another exemplary plot of the effect of various types of cap liners on slip torque, according to one embodiment.

(72) FIG. 27 illustrates another exemplary plot of the effect of various types of cap liners on break torque, according to one embodiment.

(73) FIG. 28 illustrates another exemplary plot of the effect of various types of cap liners on secure seal test (SST), according to one embodiment.

(74) FIG. 29 illustrates another exemplary plot of the effect of various types of cap liners on oxygen transmission rate (OTR), according to one embodiment.

(75) FIG. 30 illustrates another exemplary plot of the effect of various types of cap liners on slip torque, according to one embodiment.

(76) FIG. 31 illustrates another exemplary plot of the effect of various types of cap liners on break torque, according to one embodiment.

(77) FIG. 32 illustrates another exemplary plot of the effect of various types of cap liners on secure seal test (SST), according to one embodiment.

(78) FIG. 33 illustrates another exemplary plot of the effect of various types of cap liners on oxygen transmission rate (OTR), according to one embodiment.

(79) FIG. 34 illustrates an exemplary diagram of cap settings that are adjusted for the proper application of a cap closure to a bottle, according to one embodiment.

(80) FIGS. 35-36 illustrate exemplary physical and chemical properties of OXYLON® 420 liner material, according to one embodiment.

(81) FIGS. 37-38 illustrate exemplary physical and chemical properties of OXYLON® 907 liner material, according to one embodiment.

(82) FIGS. 39-40 illustrate exemplary physical and chemical properties of OXYLON® CS25 liner material, according to one embodiment.

DETAILED DESCRIPTION

(83) A system and method for implementing cap closure for a carbonated beverage is disclosed. According to one embodiment, an apparatus includes a cap liner having a circular ring shape. The apparatus further comprises an outer lip and an inner portion of the circular ring shape. The outer lip is wider than the inner portion, and the outer lip has two or more members extending away from a center of the outer lip.

(84) Each of the features and teachings disclosed herein can be utilized separately or in conjunction with other features and teachings to provide a system and method for implementing cap closure for carbonated and oxygen sensitive wine. Representative examples utilizing many of these additional features and teachings, both separately and in combination, are described in further detail with reference to the attached figures. This detailed description is merely intended to teach a person of skill in the art further details for practicing preferred aspects of the present teachings and is not intended to limit the scope of the claims. Therefore, combinations of features disclosed above in the detailed description may not be necessary to practice the teachings in the broadest sense, and are instead taught merely to describe particularly representative examples of the present teachings.

(85) In the description below, for purposes of explanation only, specific nomenclature is set forth to provide a thorough understanding of the present disclosure. However, it will be apparent to one

skilled in the art that these specific details are not required to practice the teachings of the present disclosure.

(86) Moreover, the various features of the representative examples and the dependent claims may be combined in ways that are not specifically and explicitly enumerated in order to provide additional useful embodiments of the present teachings. It is also expressly noted that all value ranges or indications of groups of entities disclose every possible intermediate value or intermediate entity for the purpose of original disclosure, as well as for the purpose of restricting the claimed subject matter. It is also expressly noted that the dimensions and the shapes of the components shown in the figures are designed to help to understand how the present teachings are practiced, but not intended to limit the dimensions and the shapes shown in the examples.

(87) According to one embodiment, the present system and method provides a cap liner configuration that is formed using injection molding for an aluminum 30 mm diameter by 60 mm tall (30×60) screw-cap closure. The present liner configuration is determined using a design-of-experiment (DOE) methodology.

(88) The present system and method provides a cap (e.g., an aluminum screw-cap) that includes a cap liner with a specified liner profile. The present cap provides a sealing performance that is controlled largely based on its liner characteristics including the liner's components and the liner's physical profile. The present system and method provides a cap liner that seals sufficiently to prevent the beverage from leaking out of the package. The present system and method further provides a cap liner that controls the transmission of oxygen from the air outside the package into the product. The amount of oxygen allowed into the package along with the rate of oxygen transmission can have a significant impact on the beverage's nutritional content, flavor and mouth feel. Liner types have traditionally been chosen by cap manufacturers (e.g., G3) with a focus on ease of use, performance, and price. According to one embodiment, the present cap liner provides an OTR value close to that of a SARANEX™ lined cap (0.0008 cc O.sub.2/24 hours) and holds an internal package pressure greater than 70 psi.

(89) FIGS. 1(a)-1(c) illustrate respectively a top view, a cross-sectional view, and a detailed cross-sectional view of an exemplary cap liner designed for use in a 30 mm diameter aluminum top-side seal closure, according to one embodiment. The cap liner **100** is of a circular disc shape with a diameter of 28.84 millimeters (mm). The cap liner **100** includes two circular-shaped cavities **101** and **102** that are each depressed into both sides of the cap liner **100**. The two circular-shaped cavities **101** and **102** create an inner ring **105** (“inner rib”), within the cap liner **100**. The inner ring **105** has an inner rib height **107** of 1.00 mm created by the depression of either circular-shaped cavity **101** and **102**.

(90) The cap liner **100** further includes two ring-shaped troughs **103** and **104** that are located outside the two circular-shaped cavities **101** and **102** and inside the outer circumference of the cap liner **100**. Each ring-shaped trough **103** and **104** is depressed into both sides of the cap liner **100**. The base of the ring for each ring-shaped trough **103** and **104** is 5.00 mm wide. The two ring-shaped troughs **103** and **104** create an outer ring (“outer lip”) **106** directly adjacent to the circumference of the cap liner **100**. Each ring-shaped trough **103** and **104** is depressed at a height of 0.50 mm relative to the cap liner **100** trough's base and outer ring peak. The outer ring **106** has an overall outer lip height **108** of 1.75 mm and an outer lip angle **109** of 22° 15'. The outer ring **106** further includes an outer nub **110** designed to help retain the liner in the aluminum screw cap.

(91) FIGS. 2(a)-2(c) illustrate respectively a top view, a cross-sectional view, and a detailed cross-sectional view of another exemplary cap liner designed for use in a 30 mm diameter aluminum top-side seal closure, according to one embodiment. The cap liner **200** is of a circular disc shape with a diameter of 28.84 mm. The cap liner **200** includes two circular-shaped cavities **201** and **202** that are each depressed into both sides of the cap liner **200**. The two circular-shaped cavities **201** and **202** create an inner ring **205** (“inner rib”) within the cap liner **200**. The inner ring **205** has an inner rib height **207** of 1.00 mm created by the depression of either circular-shaped cavity **201** and **202**.

(92) The cap liner **200** further includes two ring-shaped troughs **203** and **204** that are located outside the two circular-shaped cavities **201** and **202** and inside the outer circumference of the cap liner **200**. Each ring-shaped trough **203** and **204** is depressed into both sides of the cap liner **200**. The base of the ring for each ring-shaped trough **203** and **204** is 5.00 mm wide. The two ring-shaped troughs **203** and **204** create an outer ring **206** (“outer lip”) directly adjacent to the circumference of the cap liner **200**. Each ring-shaped trough **203** and **204** is depressed at a height of 0.25 mm relative to the cap liner **200** trough's base and outer ring peak. The outer ring **206** has an overall outer lip height **208** of 1.25 mm and an outer lip angle **209** of 22° 15'. The outer ring **206** further includes an outer nub **210** designed to help retain the liner in the aluminum screw cap.

(93) FIGS. 3(a)-3(c) illustrate respectively a top view, a cross-sectional view, and a detailed cross-sectional view of another exemplary cap liner designed for use in a 30 mm diameter aluminum top-side seal closure, according to one embodiment. The cap liner **300** is of a circular disc shape with a diameter of 28.84 mm. The cap liner **300** includes two circular-shaped cavities **301** and **302** that are each depressed into both sides of the cap liner **300**. The two circular-shaped cavities **301** and **302** create an inner ring **305** (“inner rib”) within the cap liner **300**. The inner ring **305** has an inner rib height **307** of 0.50 mm created by the depression of either circular-shaped cavity **301** and **302**.

(94) The cap liner **300** further includes two ring-shaped troughs **303** and **304** that are located outside the two circular-shaped cavities **301** and **302** and inside the outer circumference of the cap liner **300**. Each ring-shaped trough **303** and **304** is depressed into both sides of the cap liner **300**. The base of the ring for each ring-shaped trough **303** and **304** is 5.00 mm wide. The two ring-shaped troughs **303** and **304** create an outer ring **306** (“outer lip”) directly adjacent to the circumference of the cap liner **300**. Each ring-shaped trough **303** and **304** is depressed at a height of 0.50 mm relative to the cap liner **300** trough's base and outer ring peak. The outer ring **306** has an overall outer lip height **308** of 1.75 mm and an outer lip angle **309** of 22° 15'. The outer ring **306** further includes an outer nub **310** designed to help retain the liner in the aluminum screw cap.

(95) FIGS. 4(a)-4(c) illustrate respectively a top view, a cross-sectional view, and a detailed cross-sectional view of another exemplary cap liner designed for use in a 30 mm diameter aluminum top-side seal closure, according to one embodiment. The cap liner **400** is of a circular disc shape with a diameter of 28.84 mm. The cap liner **400** includes two circular-shaped cavities **401** and **402** that are each depressed into both sides of the cap liner **400**. Each circular-shaped cavity **401** and **402** has a base diameter of 25.75 mm. The two circular-shaped cavities **401** and **402** create an outer ring **403** (“outer lip”) directly adjacent to a circumference of the cap liner **400**. The outer lip **403** has a height **405** of 1.51 mm and an outer lip angle **406** of 22° 15'. The outer ring **403** further includes an outer nub **404** designed to help retain the liner in the aluminum screw cap.

(96) FIGS. 5(a)-5(c) illustrate respectively a top view, a cross-sectional view, and a detailed cross-sectional view of another cap liner designed for use in a 30 mm diameter aluminum top-side seal closure, according to one embodiment. The cap liner **500** is of a circular disc shape with a diameter of 28.84 mm. The cap liner **500** includes two circular-shaped cavities **501** and **502** that are each depressed into both sides of the cap liner **500**. Each circular-shaped cavity **501** and **502** has a base diameter of 25.75 mm. The two circular-shaped cavities **501** and **502** create an outer ring **503** (“outer lip”) directly adjacent to a circumference of the cap liner **500**. The outer lip **503** has a height **505** of 1.25 mm and an outer lip angle **506** of 22° 15'. The outer ring **503** further includes an outer nub **504** designed to help retain the liner in the aluminum screw cap.

(97) FIGS. 6(a)-6(c) illustrate respectively a top view, a cross-sectional view, and a detailed cross-sectional view of another exemplary cap liner designed for use in a 30 mm diameter aluminum top-side seal closure, according to one embodiment. The cap liner **600** is of a circular disc shape with a diameter of 28.84 mm. The cap liner **600** includes two circular-shaped cavities **601** and **602** that are each depressed into both sides of the cap liner **600**. Each circular-shaped cavity **601** and **602** has a base diameter of 25.75 mm. The two circular-shaped cavities **601** and **602** create an outer ring **603** (“outer lip”) directly adjacent to a circumference of the cap liner **600**. The outer lip **603** has a

height **605** of 1.75 mm and an outer lip angle **606** of $22^{\circ} 15'$. The outer ring **603** further includes an outer nub **604** designed to help retain the liner in the aluminum screw cap. The cap liner **600** has a similar liner profile to the cap liner **400** but with a modified outer lip height **605**.

(98) FIGS. 7(a)-7(c) illustrate respectively a top view, a cross-sectional view, and a detailed cross-sectional view of another exemplary cap liner designed for use in a 30 mm diameter aluminum top-side seal closure, according to one embodiment. The cap liner **700** is of a circular disc shape with a diameter of 28.84 mm. The cap liner **700** includes two circular-shaped cavities **701** and **702** that are each depressed into both sides of the cap liner **700**. Each circular-shaped cavity **701** and **702** has a diameter of 25.75 mm. The two circular-shaped cavities **701** and **702** create an outer ring **703** (“outer lip”) directly adjacent to a circumference of the cap liner **700**. The outer lip **703** has a height **705** of 1.63 mm and an outer lip angle **706** of $22^{\circ} 15'$. The outer ring **703** further includes an outer nub **704** designed to help retain the liner in the aluminum screw cap. The cap liner **700** has a similar liner profile to the cap liner **400** but with a modified outer lip height **705**.

(99) FIGS. 8(a)-8(c) illustrate respectively a top view, a cross-sectional view, and a detailed cross-sectional view of another exemplary cap liner designed for use in a 30 mm diameter aluminum top-side seal closure, according to one embodiment. The cap liner **800** is of a circular disc shape with a diameter of 28.84 mm. The cap liner **800** includes two circular-shaped cavities **801** and **802** that are each depressed into both sides of the cap liner **800**. Each circular-shaped cavity **801** and **802** has a diameter of 26.73 mm. The two circular-shaped cavities **801** and **802** create an outer ring **803** (“outer lip”) directly adjacent to a circumference of the cap liner **800**. The outer lip **803** has a height **805** of 1.63 mm and an outer lip angle **806** of $37^{\circ} 15'$. The outer ring **803** further includes an outer nub **804** designed to help retain the liner in the aluminum screw cap. The cap liner **800** has a similar liner profile to the cap liner **400** but with a modified outer lip height **805** and a modified outer lip angle **806**.

(100) FIGS. 9(a)-9(b) illustrate a cross-sectional view and a detailed cross-sectional view of another exemplary cap liner in a screw cap as, according to one embodiment. FIGS. 9(c)-9(e) illustrate respectively a top view, a cross-sectional view, and a detailed cross-sectional view of another exemplary cap liner as illustrated in FIG. 9(b), designed for use in a 30 mm diameter aluminum top-side seal closure, according to one embodiment. The cap liner **900** is of a circular disc shape with a diameter of 28.44 mm. The cap liner **900** includes two circular-shaped cavities **901** and **902** that are each depressed into both sides of the cap liner **900**. Each circular-shaped cavity **901** and **902** has a diameter of 25.75 mm. The two circular-shaped cavities **901** and **902** create an outer ring **903** (“outer lip”) directly adjacent to a circumference of the cap liner **900**. The outer lip **903** has a height **905** of 1.63 mm. The cap liner **900** does not include an outer nub allowing the cap liner **900** to be more easily inserted into an aluminum cap shell while still maintaining enough of an interference fit to keep it solidly retained in the finished cap.

(101) FIG. 10 (a) illustrates a detailed cross-sectional view of an “outer lip” **1000** of cap liners for a design-of experiment. FIG. 10(a) further illustrates in details a thickness **1001** of the outer lip structure **1000**, an outer lip angle **1002** measured with respect to a reference line (“datum”) and a diameter **1003** of the outer lip structure **1000** measured from a baseline relative to the other liners in the study.

(102) FIG. 10 (b) illustrates an exemplary table **1004** of the outer lip samples used in a test, according to one embodiment. The table **1004** illustrates the outer lip angle (“lip structure angle”) **1002** measured with respect to a reference line (“datum”), the thickness **1001** of the outer lip structure (“lip structure thickness”) and the diameter **1003** of the outer lip measured relative to the liners in the study, for seven different liner outer lip profiles used in the test. FIGS. 11-17 illustrate these seven test samples in detail.

(103) FIGS. 11(a)-11(c) illustrate respectively a top view, a cross-sectional view, and a detailed cross-sectional view of another exemplary cap liner designed for use in a 30 mm diameter aluminum top-side seal closure, according to one embodiment. The cap liner **1100** is of a circular

disc shape with a diameter of 1.120 in. The cap liner **1100** includes two circular-shaped cavities **1101** and **1102** that are each depressed into both sides of the cap liner **1100**. The two circular-shaped cavities **1101** and **1102** create an outer ring **1103** (“outer lip”) directly adjacent to a circumference of the cap liner **1100**. The outer lip **1103** has an overall height **1104** of 0.062 in, lip structure thickness **1105** of 0.012 in and an outer lip angle of 0°.

(104) FIGS. **12(a)-12(c)** illustrate respectively a top view, a cross-sectional view, and a detailed cross-sectional view of another exemplary cap liner designed for use in a 30 mm diameter aluminum top-side seal closure, according to one embodiment. The cap liner **1200** is of a circular disc shape with a diameter of 1.120 in. The cap liner **1200** includes two circular-shaped cavities **1201** and **1202** that are each depressed into both sides of the cap liner **1200**. The two circular-shaped cavities **1201** and **1202** create an outer ring **1203** (“outer lip”) directly adjacent to a circumference of the cap liner **1200**. The outer lip **1203** has an overall height **1204** of 0.062 in, lip structure thickness **1205** of 0.016 in and an outer lip angle of 0°.

(105) FIGS. **13(a)-13(c)** illustrate respectively a top view, a cross-sectional view, and a detailed cross-sectional view of another exemplary cap liner designed for use in a 30 mm diameter aluminum top-side seal closure, according to one embodiment. The cap liner **1300** is of a circular disc shape with a diameter of 1.120 in. The cap liner **1300** includes two circular-shaped cavities **1301** and **1302** that are each depressed into both sides of the cap liner **1300**. The two circular-shaped cavities **1301** and **1302** create an outer ring **1303** (“outer lip”) directly adjacent to a circumference of the cap liner **1300**. The outer lip **1303** has an overall height **1304** of 0.063 in, lip structure thickness **1305** of 0.008 in and an outer lip angle **1306** of 140°.

(106) FIGS. **14(a)-14(c)** illustrate respectively a top view, a cross-sectional view, and a detailed cross-sectional view of another exemplary cap liner designed for use in a 30 mm diameter aluminum top-side seal closure, according to one embodiment. The cap liner **1400** is of a circular disc shape with a diameter of 1.120 in. The cap liner **1400** includes two circular-shaped cavities **1401** and **1402** that are each depressed into both sides of the cap liner **1400**. The two circular-shaped cavities **1401** and **1402** create an outer ring **1403** (“outer lip”) directly adjacent to a circumference of the cap liner **1400**. The outer lip **1403** has a height **1404** of 0.066 in, thickness **1405** of 0.012 in and an outer lip angle **1406** of 70°. Due to the outer diameter reduction of this liner, created by the designed experiment's factor levels, an outer “nub” **1407** was added to the outer lip **1403** to help retain the liner in the aluminum screw cap.

(107) FIGS. **15(a)-15(c)** illustrate respectively a top view, a cross-sectional view, and a detailed cross-sectional view of another exemplary cap liner designed for use in a 30 mm diameter aluminum top-side seal closure, according to one embodiment. The cap liner **1500** is of a circular disc shape with a diameter of 1.120 in. The cap liner **1500** includes two circular-shaped cavities **1501** and **1502** that are each depressed into both sides of the cap liner **1500**. The two circular-shaped cavities **1501** and **1502** create an outer ring **1503** (“outer lip”) directly adjacent to a circumference of the cap liner **1500**. The outer lip **1503** has an overall height **1504** of 0.066 in, lip structure thickness **1505** of 0.008 in and an outer lip angle **1506** of 35°. Due to the outer diameter reduction of this liner, created by the designed experiment's factor levels, an outer “nub” **1507** was added to the outer lip **1503** to help retain the liner in the aluminum screw cap.

(108) FIGS. **16(a)-16(c)** illustrate respectively a top view, a cross-sectional view, and a detailed cross-sectional view of another exemplary cap liner designed for use in a 30 mm diameter aluminum top-side seal closure, according to one embodiment. The cap liner **1600** is of a circular disc shape with a diameter of 1.120 in. The cap liner **1600** includes two circular-shaped cavities **1601** and **1602** that are each depressed into both sides of the cap liner **1600**. The two circular-shaped cavities **1601** and **1602** create an outer ring **1603** (“outer lip”) directly adjacent to a circumference of the cap liner **1600**. The outer lip **1603** has an overall height **1604** of 0.065 in, lip structure thickness **1605** of 0.016 in and an outer lip angle **1606** of 105°. Due to the outer diameter reduction of this liner, created by the designed experiment's factor levels, an outer “nub” **1607** was

added to the outer lip **1603** to help retain the liner in the aluminum screw cap.

(109) FIGS. **17(a)**-**17(c)** illustrate respectively a top view, a cross-sectional view, and a detailed cross-sectional view of another exemplary cap liner designed for use in a 30 mm diameter aluminum top-side seal closure, according to one embodiment. The cap liner **1700** is of a circular disc shape with a diameter of 1.120 in. The cap liner **1700** includes two circular-shaped cavities **1701** and **1702** that are each depressed into both sides of the cap liner **1700**. The two circular-shaped cavities **1701** and **1702** create an outer ring **1703** ("outer lip") directly adjacent to a circumference of the cap liner **1700**. The outer lip **1703** has an overall height **1704** of 0.063 in, lip structure thickness **1705** of 0.012 in and an outer lip angle **1706** of 140°. Due to the outer diameter reduction of this liner, created by the designed experiment's factor levels, an outer "nub" **1707** was added to the outer lip **1703** to help retain the liner in the aluminum screw cap.

(110) FIGS. **18(a)**-**18(c)** illustrate respectively a top view, a cross-sectional view, and a detailed cross-sectional view of another exemplary cap liner designed for use in a 30 mm diameter aluminum top-side seal closure, according to one embodiment. The cap liner **1800** is of a circular disc shape with a diameter of 1.120 in. The cap liner **1800** includes two circular-shaped cavities **1801** and **1802** that are each depressed into both sides of the cap liner **1800**. Each circular-shaped cavity **1801** and **1802** has a diameter of 1.010 in. The two circular-shaped cavities **1801** and **1802** create an outer ring **1803** ("outer lip") directly adjacent to a circumference of the cap liner **1800**. The outer lip **1803** has a height **1804** of 0.061 in and an outer lip angle **1805** of 22.25°. Dual injection molding technique is used to create the outer lip **1803** or portions thereof with a different material than the remainder of the liner. For example, low density polyethylene (LDPE) may be used to create the inner portions of the liner **1800**, and thermoplastic elastomers (TPE) may be over-molded to the LDPE to form the liner's outer edge including the entire outer lip **1803** or just portions of the outer lip. The thickness of the over-molded TPE material can vary during the molding process depending on the amount of TPE required to obtain the desired effect. For example, as shown in FIG. **18(b)**, the thickness of the over-molded portion **1807** on the outside edge of the liner **1800** can be 0.018 in, 0.039 in, 0.069 in cross-sectional width or the like, depending on the amount of TPE desired in the final liner design. The cap liner **1800** does not include an outer nub allowing the cap liner **1800** to be more easily inserted into an aluminum cap shell while still maintaining enough of an interference fit to keep it solidly retained in the finished cap.

(111) FIGS. **19(a)**-**19(c)** illustrate respectively a top view, a cross-sectional view, and a detailed cross-sectional view of another exemplary cap liner designed for use in a 30 mm diameter aluminum top-side seal closure, according to one embodiment. In this exemplary embodiment, the cap liner's **1900** center is removed to save material and therefore reduce the manufacturing cost. The cap liner **1900** is consequently of a circular ring shape with a diameter of 1.120 in. The cap liner **1900** includes two annular surfaces **1901** and **1902**. The annular surfaces **1901** and **1902** are each depressed into both sides of the cap liner **1900** ring and create an outer lip **1903** directly adjacent to a circumference of the cap liner **1900**. The outer lip **1903** includes an inner portion **1904** and an outer portion **1905**. Dual injection molding technique is used to create the annular surfaces **1901** and **1902** and an over-mold **1906** creating the outer portion **1905** of the outer lip **1903**. The material used to create the outer portion of the outer lip **1905** may encroach on the inner portion of the outer lip **1904** depending on the liner's functional requirements. Low density polyethylene (LDPE), polypropylene (PP), or any material providing the appropriate properties, such as rigidity, is used for inner portion **1904** of the outer lip **1903** formed by the annular surfaces **1901** and **1902**, and a thermoplastic elastomer (TPE), polyethylene terephthalate (PET), high-density polyethylene (HDPE), polyamide or other material providing appropriate properties, such as reduced oxygen transmission, is used for the over-mold **1906** portion of the cap liner **1900** including all or some of the outer lip **1903**. The outer lip **1903** may be shaped as necessary to provide for the proper performance characteristics. The triangular shape of the outer lip **1903** is designed to help to

provide a larger surface area and a better LDPE to TPE adhesive strength to maintain the liner's integrity.

(112) FIGS. **20(a)-20(b)** illustrate a cross-sectional view and a detailed cross-sectional view of another exemplary cap liner in a screw cap as, according to one embodiment. FIGS. **20(c)-20(f)** illustrate respectively a top view, a 3-dimensional view, a cross-sectional view, and a detailed cross-sectional view of another exemplary cap liner as illustrated in FIG. **20 (b)**, designed for use in a 30 mm diameter aluminum top-side seal closure, according to one embodiment. The cap liner **2000** is of a circular disc shape with a diameter of 1.049 in. The cap liner **2000** includes two circular-shaped cavities **2001** and **2002** that are each depressed into both sides of the cap liner **2000**. The two circular-shaped cavities **2001** and **2002** create an outer ring **2003** ("outer lip") directly adjacent to a circumference of the cap liner **2000**. The outer lip **2003** has a height **2004** of 0.066 in, thickness **2005** of 0.008 in and an outer lip angle **2006** of 51.79°. The outer lip **2003** further includes three tabs **2007**, **2008**, **2009**, that are created by removing extra material from the outer lip **2003** nub to help reduce the amount of material and cost. The three tabs **2007**, **2008**, **2009** provide enough of an interference fit to keep the liner solidly retained in the finished cap.

(113) FIG. **21** illustrates an exemplary graph for determining a maximum internal package pressure in a rigid container filled to the appropriate volume, according to one embodiment. A plot **2100** illustrates that for a maximum wine temperature of about 110° F. and a carbon dioxide concentration in the beverage of about 5.4 gram/liter, the expected maximum product pressure is 70 psi.

(114) FIGS. **22-25** illustrate exemplary plots of the effect of various types of cap liners on slip torque, break torque, secure seal test (SST), and oxygen transmission rate (OTR) respectively, according to one embodiment. In particular, FIGS. **22-25** illustrate plots of the effect of the cap liner's material and physical profiles LDPE-1, LDPE-2, LDPE-3, LDPE-4, LDPE-5, Oxyton 420-1, Oxyton 420-2, Oxyton 420-3, Oxyton 420-4, Oxyton 420-5 on slip torque, break torque, SST, and OTR, respectively.

(115) Referring to FIG. **22**, the slip torque distributions for cap liners made of a TPE called OXYLON® 420 (e.g., Oxyton 420-3, Oxyton 420-4, and Oxyton 420-5) indicate higher slip torques than the cap liners made of a LDPE liner material (e.g., LDPE-1, LDPE-2, LDPE-3, LDPE-4, and LDPE-5). According to one embodiment, adjustments to the slip torque can be made by using a slip agent. The slip agent can be based upon any of a number of materials approved for food contact that are added to the liner material of a liner profile to reduce slip torque. These slip agents include, but are not limited to, amides, erucamide, oleamide, polyethylene beads, lanolin, and carnauba waxes.

(116) Referring to FIG. **23**, the break torque scatter plots for cap liners made of an OXYLON® liner material (e.g., Oxyton 420-3, Oxyton 420-4, and Oxyton 420-5) indicate only slight differences when compared with the cap liners made of a LDPE liner material (e.g., LDPE-1, LDPE-2, LDPE-3, LDPE-4, and LDPE-5).

(117) Further, referring to FIG. **24**, the cap liners made of an OXYLON® 420 liner material (Oxyton 420-3, Oxyton 420-4, and Oxyton 420-5) and cap liners made of LDPE (LDPE-3, and LDPE-4) performed well in the SST test by holding the targeted minimum SST of 150 psi for each of 3 pressure cycles. However, the cap liners made of an OXYLON® 420 liner material (Oxyton 420-1, and Oxyton 420-2) and a LDPE liner material (LDPE-1, LDPE-2, and LDPE-5) did not hold pressure well with scatter plots showing a number of the samples leaking below the targeted minimum SST of 150 psi.

(118) Referring to FIG. **25**, the OTR values for the cap liners made of OXYLON® 420 liner material (Oxyton 420-1, Oxyton 420-2, Oxyton 420-3, Oxyton 420-4, and Oxyton 420-5) are lower than the OTR values for the cap liners made of a LDPE liner material (LDPE-1, LDPE-2, LDPE-3, LDPE-4, and LDPE-5). In particular, the OTR for the cap liners made of an OXYLON® 420 liner material is about half of the OTR for the cap liners made of an LDPE liner material.

(119) A cap liner profile is selected based on a targeted slip and a targeted SST. For example, the targeted slip is 6-20 in-lbs., and the targeted SST is a minimum of 150 psi. Referring to FIGS. 22 and 24, the cap liner LDPE-4 provides the best results for a targeted slip of 10-20 in-lbs. and a targeted minimum SST of 150 psi. The cap liner LDPE-3 provides subsequent closest results for the targeted slip and the targeted SST, followed by the cap liner LDPE-2.

(120) FIGS. 26-29 illustrate exemplary plots of slip torque, break torque, SST, and OTR test results for various cap liners respectively, according to one embodiment. In particular, FIGS. 26-29 illustrate plots of slip torque test results, break torque test results, SST test results, and OTR test results respectively for the cap liners LDPE-6, LDPE-7, LDPE-8, Oxylon 907-4 and Oxylon CS25-4.

(121) According to one embodiment, a slip agent chosen from a number of materials approved for food contact are added to the liner material of a liner profile to reduce slip torque. These slip agents include, but are not limited to, amides, erucamide, oleamide, polyethylene beads, lanolin, and carnauba waxes.

(122) Referring to FIG. 26 and FIG. 28, the cap liner made of LDPE-7 liner material has a lowest average slip torque of 14.08 in-lbs. and an average SST of 150 psi. Referring to FIG. 29, the average OTR of the cap liner LDPE-8 is the highest compared to the average OTR of the other cap liners LDPE-6 and LDPE-7. These liners LDPE-6, LDPE-7, LDPE-8 are all made using the same LDPE material and differ only in their physical profile.

(123) Referring to FIG. 26, the cap liner made of using OXYLON® CS25 liner material (Oxylon CS25-4) has a lower average slip than the cap liner made of an OXYLON® 907 liner material (Oxylon 907-4). However, both Oxylon CS25-4 and Oxylon 907-4 have the same average SST of 150 psi, as illustrated in FIG. 28. Oxylon 907-4 has a lower OTR than Oxylon CS25-4, as illustrated in FIG. 29. These liners Oxylon 907-4, Oxylon CS25-4 are made using the same physical profile and differ only in the TPE materials.

(124) FIGS. 30-33 illustrate exemplary plots of slip torque, break torque, SST, and OTR test results for various cap liners respectively, according to one embodiment. In particular, FIGS. 30-33 illustrate plots of slip torque test results, break torque test results, SST test results, and OTR test results respectively for the cap liners made of, LDPE used for the exemplary embodiments of FIGS. 9(a)-9(c), LDPE-OL1, LDPE-OL2, LDPE-OL3, LDPE-OL4, LDPE-OL5, LDPE-OL6 and LDPE-OL7.

(125) According to one embodiment, a slip agent chosen from a number of materials approved for food contact are added to the liner material of a liner profile to reduce slip torque. These slip agents include, but are not limited to, amides, erucamide, oleamide, polyethylene beads, lanolin, and carnauba waxes.

(126) Referring to FIG. 30, the cap liner made of LDPE-OL1 liner material has a lowest average slip torque of approximately 7 in-lbs. However, referring to FIG. 32, the cap liners having specific profiles LDPE—FIG. 9(a)(b)(c) and LDPE-OL3 all held an average SST pressure of 150 psi. Referring to FIG. 33, the average OTR of the cap liner LDPE-OL1 is the highest compared to the average OTR of the other cap liners LDPE—FIG. 9(a)(b)(c), LDPE-OL2, LDPE-OL3, LDPE-OL4, LDPE-OL5, LDPE-OL6 and LDPE-OL7. All liners LDPE—FIG. 9(a)(b)(c), LDPE-OL1, LDPE-OL2, LDPE-OL3, LDPE-OL4, LDPE-OL5, LDPE-OL6 and LDPE-OL7 were made using the same LDPE material and differed only in their physical profiles.

(127) According to one embodiment, each cap liner sample is manually inserted into a 30×60 mm aluminum screw cap and applied to a wine bottle. (U.S. standard-GPI finish 1680). The wine bottle may contain any type of liquid, such as water or lightly carbonated wine. In one embodiment, each wine bottle is pressurized to a desired level to a mimic a desired carbon dioxide level (e.g., 2-4 g CO.sub.2/L) within the wine bottle. Each cap closure may be applied to a desired bottle finish using any capper equipment known in the art.

(128) According to one embodiment, the capper settings used to apply each cap closure containing

each liner Oxyton 420-1, Oxyton 420-2, Oxyton 420-3, Oxyton 420-4, Oxyton 420-5, LDPE-1, LDPE-2, LDPE-3, LDPE-4, LDPE-5 are: a top-load of 450 lbf; a reform depth of 0.070 inches; a thread roller force of 28 lbf; a pilfer roller force of 20 lbf; thread and pilfer roller heights are set to proper gage.

(129) According to another embodiment, the cap setting used to apply each cap closure containing each liner profile Oxyton 907-4, Oxyton CS25-4, LDPE-6, LDPE-7, LDPE-8, are: a top load of 400 lbf; a reform depth of 0.074 inches, a thread roller force of 28 lbf; a pilfer roller force of 25 lbf; and roller heights are set to proper gage.

(130) FIG. 34 illustrates an exemplary diagram of cap settings used to apply a cap closure to a bottle, according to one embodiment. A top load 3303 refers to an amount of force that is applied to the top of a cap metal 3301 to compress the cap liner within the cap metal 3301 onto a bottle's sealing surface 3302. A reform depth 3304 refers to a depth that the cap metal 3301 is forced tight over the bottle's seal surface 3302. A thread roller 3304 pushes on a side of the cap metal 3301 to apply a thread roller force which forms threads in the cap metal 3301 during application.

(131) According to one embodiment, the cap liner samples are allowed to sit for a minimum of 24 hours prior to testing. Each cap liner sample is evaluated for pressure retention using a SECUREPAK® (a type of package testing instrument) Secure Seal Tester by pressurizing each sample's headspace to 150 psi, holding the pressure of each sample at 150 psi for 5-10 seconds and then releasing the pressure to 0 psi. This pressurization is repeated three times for each sample, with the maximum pressure (psi) held being recorded. In another embodiment, each cap liner sample is evaluated for oxygen transmission rate (OTR) to determine the rate at which oxygen permeates through each cap liner. The cap liner samples are allowed to equilibrate for a minimum of three days on an equilibration rack, transferred to the MOCON® OX-TRAN® 2121 testing stations and tested a minimum of three times or until the steady-state OTR is reached.

(132) Competitive cap products (ERBEN® ASTRO and GUALA® MOSS) contain injection molded liners with inner ribs. Competitors claim this inner rib helps the package retain pressure generated by the carbonated beverage. In particular, the ERBEN® ASTRO liner is patented in Italy by Strocchio et al. (Patent No. 0001378221 entitled "Plastic Gasket with a Pressure Seal for Closing Bottles or Similar Containers") and contains references to the inner rib as being integral to the liner's sealing characteristics. However, the results of the present DOE modeling illustrates that the present liner profile does not require an inner rib to increase the ability of the liner to seal. In particular, the present liner profile does not include an inner rib since the sealing performance of the liner is based on configuration of the outer lip and the material used to create the outer lip.

(133) According to one embodiment, the present cap liner profiles are injection molded, single-layered liners made from either an LDPE liner material or a TPE liner material that provides superior oxygen barrier properties. The TPE liner material may include, but not limited to, OXYLON® 420, OXYLON® 907, and OXYLON® CS25 manufactured by ACTEGA® (a company that manufactures coatings and sealants).

(134) According to another embodiment, the present cap liner profiles are dual-injection molded, double-layered liners made from an LDPE liner material and a TPE liner material that provides superior oxygen barrier properties. The TPE liner material may include, but not limited to, OXYLON® 420, OXYLON® 907, and OXYLON® CS25 manufactured by ACTEGA® (a company that manufactures coatings and sealants).

(135) FIGS. 35-36 illustrate exemplary physical and chemical properties of OXYLON® 420 liner material, according to one embodiment. FIGS. 37-38 illustrate exemplary physical and chemical properties of OXYLON® 907 liner material, according to one embodiment. FIGS. 39-40 illustrate exemplary physical and chemical properties of OXYLON® CS25 liner material, according to one embodiment.

(136) The above example embodiments have been described hereinabove to illustrate various embodiments of providing a system and method for implementing cap closure for carbonated wine.

Various modifications and departures from the disclosed example embodiments will occur to those having ordinary skill in the art. The subject matter that is intended to be within the scope of the disclosure is set forth in the following claims.

Claims

1. An apparatus, comprising: a cap liner comprising a circular disc shape comprising an inner ring and an outer ring, wherein the outer ring comprises a substantially flat outer edge extending between top and bottom surfaces of the outer ring, the cap liner further comprising a protuberant outer nub projecting radially outward from the substantially flat outer edge, wherein the protuberant outer nub is substantially centered between the top and bottom surfaces of the outer ring without extending to the top and bottom surfaces of the outer ring, wherein the circular disc shape defines two ring-shaped troughs disposed between the inner ring and the outer ring, wherein the cap liner further comprises a center disc disposed within the inner ring, wherein the center disc is planar and extends to a center of the cap liner from the inner ring, wherein the protuberant outer nub defines an outer perimeter of the cap liner, and wherein a cross-section of the protuberant outer nub has a convex outer surface that is rounded outward.
 2. The apparatus of claim 1, wherein the cap liner comprises low-density polyethylene (LDPE).
 3. The apparatus of claim 1, wherein the cap liner comprises thermoplastic elastomer (TPE).
 4. The apparatus of claim 1, wherein at least a portion of the cap liner is injection molded or dual injection molded.
 5. The apparatus of claim 1, wherein at least a portion of the cap liner is over-molded.
 6. The apparatus of claim 1, wherein the cap liner comprises a slip agent.
 7. The apparatus of claim 6, wherein the slip agent comprises one or more of an amide, erucamide, oleamide, polyethylene beads, lanolin, or a carnauba wax.
 8. The apparatus of claim 1, further comprising a cap.
 9. The apparatus of claim 8, wherein the cap liner is configured to fit inside the cap.
 10. The apparatus of claim 1, wherein the cap liner prevents oxygen from transmitting into a beverage container.
 11. The apparatus of claim 1, wherein the cap liner has a friction factor that allows a cap of a beverage container to be opened using human force.
 12. The apparatus of claim 1, wherein the cap liner maintains pressure within a beverage container having a carbonated beverage.
 13. The apparatus of claim 1, wherein the ring-shaped troughs are separated by an intermediate ring connecting the inner ring and the outer ring.
 14. The apparatus of claim 13, wherein the intermediate ring comprises a substantially uniform thickness.
 15. The apparatus of claim 1, wherein the center disc comprises a substantially uniform thickness.
 16. The apparatus of claim 1, wherein the cap liner is symmetrical about a central plane.
 17. The apparatus of claim 1, wherein the inner ring comprises an inner rib height and the outer ring comprises an outer rib height, and wherein the inner rib height is greater than the outer rib height.
 18. The apparatus of claim 16, wherein a thickness of the protuberant outer nub in a direction perpendicular to the central plane is less than a minimum thickness of the center disc.
 19. The apparatus of claim 16, wherein a thickness of the protuberant outer nub in a direction perpendicular to the central plane is less than a separation distance between the two ring-shaped troughs.
 20. The apparatus of claim 13, wherein a thickness of the intermediate ring is less than one third of a radial length of the intermediate ring.
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